Fleet Demo Post-Mortem

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I. INTRODUCTION

This demo of Fleet is a basic 3D shoot-em-up (shmup) inspired by the StarFox franchise. The player flies a starfighter after choosing a custom loadout for their ship and selecting a level to attack. The loadout screen requires the player to place shields, weapons, and batteries in the starfighter and the galaxy map requires the player to choose among strategic targets for missions. The proposal originally imagined a roguelite mechanic in which the player was a named starfighter pilot and when a ship is destroyed, they are configuring a new starfighter and controlling a new pilot. This was meant to be an in-universe explanation for the player continuing despite the destruction of the ship and the death of the pilot. This feature has not been implemented in the demo. The galaxy map is meant to allow players to strategically cripple the war-economy of an enemy star empire before attempting a final run on the capital whenever they are ready, including immediately, where the number and strength of the enemy in the "boss" level is affected by what the player has chosen to do before the attack. This was not implemented in the demo version.

II. FEATURES IMPLEMENTED

The player controls a starfigher with horizontal and vertical maneuverability, left and right knife-edge flying, barrel rolls in both directions, and brake and boost. Before flying, the player configures a loadout for their starfighter that requires them to place shield and laser modules as well as batteries. The amount and layout of each determines the behavior and equipment of the fighter. The loadout screen will force the player to choose at least one laser module, but they are free to select whatever combination of available modules after that. For the first three laser modules, the number of lasers fired by the ship is equal to the number of modules. After 3, laser strength is increased for each additional module. Shields exist after one placed shield module and are increased in strength for each additional module after that. Placed battery area must be at least equal to module area, but additional area provides additional power and batteries that are connected provid more power. There is a default loadout and the player can save up to two custom loadouts.

After configuring their starfighter, the player selects a destination level from a map which includes images for an asteroid field mining, fuel-station gas giants, moons with defense satelites, earth-like M-class planets with fighter factories, and a boss level. Currently the player can only choose the asteroid-mining level in the demo. The player only has laser weapons as placed in the loadout screen. The training level and the asteroid level are randomly generated rather than handcrafted on-rails.

Brown asteroids are rock asteroids. Silver asteroids are metal heavy asteroids and require more hits to destroy. Both types of asteroids have a chance to split into smaller asteroids. Red enemy ships are the player model recolored and will shoot at the player. Buildings in training mode have a random height and some of them have towers that will shoot at the player.

Additionally:

1) The soundtrack is my first attempt at writing music, so it's not great, but I am happy with it.
2) Sound effects are:
    a) Player laser shooting
    b) Player laser hitting obstacles
    c) Asteroids exploding
    d) Player maneuvering with jets
    e) Player getting hit by enemy lasers
    f) Player ship damaged and in danger

III. FEATURES REMOVED

1) The demo does not have a game over and there is no winning for the current state of the game.
2) There are no hand-crafted "on-rails" levels, only randomly generated ones. Only the asteroid level was implemented.
3) There is no named pilot system or counting ships. There are no roguelite elements that allow for upgrades.
4) There is no management of fleet resources. We assume infinite fuel and components.

IV. PROBLEMS ENCOUNTERED

1) Art was a huge barrier for me. I did my best with basic polygons and I am not a great 2D artist, but the very minimal art style led to my dissatisfaction with the project. I sometimes feel this while making 2D games in pico-8 but the lower expectations for 2D retro pixel games means I am happier quicker with how the game looks. I am not happy with how this game looks but making it better is beyond my skills.

2) Choosing the correct nodes and implementing 3D features was more challenging than I anticipated. For example, I still don't like how the ship flies. It is better than it was when it just moved up,down,left, right, but it feels nothing like playing StarFox and I am not sure if I am capable of improving it. I had to ask an AI how to implement features I wanted on multiple occasions.
3) Enemy fighter behavior was really frustrating. The turrets were easy because they are static and just come right at the player, but getting the enemy fighters to come on screen at the right distance and stay long enough to be shot was more difficult to balance right than I expected.
4) While playtesting I realized that I should have had the player really control the targeting reticule and have the ship follow the reticule rather than my current implementation that moves the ship and has the reticule follow, but I don't think I have time to change and fix it at this point. Definitely would need to be fixed if I continued to work on the game.
5) I need a lot more playtesting to find the right balance for costs in the loadout. I think there could be a really deep and fun gameplay in resource management for the right balance of batteries, lasers, and shields, but because I have been trying to build the levels I have not had enough time to get a feel for what costs and bonuses are correct.
6) Time management, time budget, and motivation. I think I did a good job on time management for all but the penultimate week of the project, where I crashed after finishing summer camps, but during those other weeks time budget as a solo dev was simply too small for me to accomplish the goals I originally set out to do. I was also surprised about how my motivation changed in relation to working on the game when I "had" to work on the game rather than working on it as a hobby for fun. I normally enjoy work and my hobbies both.

V. Conclusion

Despite my disappointment with the current state of the project as of the due date, I overall had a good experience. Some benefits that I can think of:

1) I finally learned how Godot works to the level I am comfortable teaching middle and high school students how to make basic games in Godot.
2) I finally forced myself to learn how to make music using pico8's music editor. It is certainly not masterful work, but I now know enough that I will be able to teach my students how to use it and let their natural talents take it from there.
3) I learned that I would probably not want to make games as a job and it is better than I continue to do it as a hobby where I can stop when I want to.
4) I have a very rough demo of an idea for a game I had planned for a while and could hire an artist to make me some art and continue to improve on it.

I love making games and teaching game design and this course was a net positive for me despite not doing as well with the game as I wanted.

Thanks for a great summer course, Dr. JJ!